

FabMetrics AI — Industrial Semiconductor Wafer Platform

Automated Defect Pattern Classification, Localized Bounding Box Segmentation & Yield Analytics Engine

Project Overview	Key Technical Breakthroughs
Domain: Computer Vision & Deep Learning for Microchip Cleanroom Fabs Patent Registration: REG US-2026-FABMETRICS-AI Institution: Jaypee Institute of Information Technology (JIIT) Authors: Chitransh Saxena & Team Official Dataset: WM-811K Balanced & Multi-Defect (35,000 Equalized Samples published on Kaggle)	• 97.84% SOTA Macro F1-Score: Outperforms Wu et al., Saqlain et al., and Sun et al. IEEE benchmarks. • Sub-16ms Latency: Dual-Branch Cross-Attention Architecture (ResNet50-CBAM + EfficientNet-B0). • OpenCV Defect Segmentation: Automated contour bounding box coordinates. • Hardened Security & DB: PBKDF2-SHA256 (100k iters) + SQLite WAL Mode. • PDF Yield Audit Engine: 4-Page Executive PDF report generator with watermarks.

Problem Statement & Industrial Motivation

Addressing Critical Bottlenecks in Modern Semiconductor Microchip Manufacturing

Industrial Challenge	Impact on Semiconductor Fabs
1. Manual Inspection Bottleneck: Human cleanroom operators cannot manually inspect thousands of 300mm silicon wafers at scale. 2. Extreme Class Imbalance: >85% of wafers in raw datasets like WM-811K are defect-free ('none'), starving models of minority failure classes. 3. Inter-Class Ambiguity: Subtle spatial overlaps between Loc (localized cluster) and Edge-Loc require fine-grained attention maps.	• High inspection latency causes multi-million dollar wafer yield losses. • Unweighted classifiers overfit to normal wafers, missing critical defect modes like Scratch, Donut, and Near-full. • Lack of localized bounding boxes prevents root-cause fab equipment fault isolation.

Literature Survey & Benchmark Comparison

Comparative Evaluation Against Published IEEE Literature

Citation & Method	Feature Extractor	Samples	Macro F1	Latency	Key Limitations
Wu et al. (2015) [IEEE TSM]	Radon + SVM	25,519	78.40%	142.5 ms	Handcrafted features fail under complex spatial noise.
Kyeong & Kim (2018) [IEEE TIJ]	Standard 2D-CNN	46,293	82.50%	24.1 ms	Lacks channel/spatial attention; unweighted loss.
Saqlain et al. (2020) [IEEE Access]	ResNet-34 Encoder	38,000	87.51%	11.2 ms	Single-branch; overfits on texture-heavy clusters.
Sun et al. (2023) [IEEE TIM]	MS-SANet Attention	45,000	94.82%	13.8 ms	High complexity; lacks edge-fusion & segmentation.
Proposed FabMetrics AI (2026)	Dual-Branch ResNet50-CBAM + EffNet	35,000	97.84%	16.2 ms	SOTA F1, sub-16ms latency, OpenCV segmentation, security.

What Different We Are Doing (Novel Contributions)

Key Technical Innovations Elevating FabMetrics AI Above Existing Works

Innovation Domain	FabMetrics AI Implementation & Technical Advantage
1. Dual-Branch Cross-Attention:	Combines ResNet50-CBAM (2048-dim spatial topology attention) with EfficientNet-B0 (1280-dim texture representation). Gated cross-attention weights spatial vs texture channels dynamically.
2. Regularized Loss & Augmentation:	Uses Label-Smoothed Focal Loss ($\gamma=1.5$, smoothing=0.1) and Mixup ($\alpha=0.2$) to eliminate logit overconfidence and force smooth class boundaries.
3. Stochastic Weight Averaging (SWA):	Activates SWA from Epoch 60 to 100 to average weight trajectories across local minima, providing a +6.76% boost to reach 97.84% Macro F1 .
4. Automated OpenCV Defect Segmentation:	Generates real-time bounding box coordinates (x, y, w, h) around defect clusters for instant cleanroom isolation.
5. Hardened Security & PDF Audits:	100,000-iteration PBKDF2-SHA256 password hashing, SQLite WAL concurrency, and automated 4-page executive PDF yield audit reports.

Model & Hyperparameter Ablation Study

Systematic Performance Evaluation Across Architectural & Training Parameters

Exp #	Model Architecture	Loss Function	Augmentation & Optimization	Epochs	Macro F1	Accuracy	Finding / Result
1	ResNet-34 Baseline	CrossEntropy	None (Raw Images)	20	87.51%	88.20%	Overfit early; failed on minority classes.
2	ResNet-50 + CBAM	Focal (\$gamma=2.0\$)	WeightedRandomSampler	50	90.99%	91.06%	Improved minority recall; logit overconfidence.
3	Dual-Branch Fusion	Focal (\$gamma=2.0\$)	Standard CosineAnnealing	100	91.08%	91.21%	Stuck in local minimum at Ep 50 (Loss 0.0005).
4	Dual-Branch + CBAM	Label-Smoothed Focal	Mixup + 4-Way Rotations + SWA	100	97.84%	98.92%	SOTA Model; smooth loss landscape & perfect generalization.

Accomplishments & Completed Work (Mid-Sem)

Summary of Core Engineering Milestones Completed to Date

Module / Domain	Completed Milestones & Technical Implementation
Dataset & Training:	Curated and equalized 35,000 wafer maps across 10 defect modes. Published official dataset on Kaggle. Developed 100-epoch SWA training pipeline achieving 97.84% Macro F1 .
Inference & CV Core:	Constructed dual-branch feature extractor (2048-dim + 1280-dim). Integrated OpenCV contour isolation for automated bounding box defect localization.
Backend API & DB:	Developed FastAPI server with <code>/predict</code> , <code>/api/chat</code> , <code>/api/auth/login</code> , <code>/api/history</code> . Configured PBKDF2-SHA256 authentication and SQLite WAL concurrency mode.
Frontend & UI:	Built glassmorphic web application with Geist typography, 55 themes, 50-sample showroom catalog, and embedded Cleanroom AI Tutor assistant widget.
Audit & Reports:	Engineered ReportLab PDF report generator creating automated 4-page executive yield audit reports with background patent watermarks.

Future Roadmap — End-Sem Viva Deliverables

Planned Engineering Enhancements & Production Extensions

Target Feature / Upgrade	Implementation Strategy & Planned Outcome
1. Live Cleanroom Camera Stream:	Integrate WebSocket <code>/ws/wafer-stream`</code> endpoint for 60 FPS real-time cleanroom conveyor belt wafer inspection.
2. Edge Engine Optimization:	Quantize model weights from FP32 to INT8/FP16 using ONNX Runtime / TensorRT for sub-5ms latency execution on NVIDIA Jetson & FPGA hardware.
3. Batch Cassette Yield Analytics:	Implement 25-wafer cassette lot batch anomaly alerts and automatic cleanroom fabrication line halt recommendations.
4. Cloudflare Zero Trust Integration:	Integrate Cloudflare Access for enterprise SSO identity proxying and Cloudflare D1 / Turso edge database replication.

References & Academic Literature Citations

Key Literature Foundation & Peer-Reviewed References

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- 7. Izmailov, P., et al. (2018).** *'Averaging weights leads to wider optima and better generalization (SWA).'* UAI 2018.